SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA | Year: I | Semester: I

Subject: Java Programming | Time: 45 mins

Date: 23 / 08 / 2023 | No. of Pages: 02

Marks: 20

Midterm Test – I (Set C)

1) Q1 is compulsory. Attempt any three questions from the remaining.
2) *Output* is mandatory for all programming questions.

Q1		**Answer briefly:**	[08]
CO1 SO-1,6 BL-2	a.	Explain Polymorphism and Inheritance.	[02]

CO1
SO-1,6
BL-3
	b.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		`class Main {`
		`  public static void main(String args[]) {`
		`  System.out.println(fun());`
		`  }`
		`  int fun() {`
		`  return 20;`
		`  }`
		`}`
			[02]

CO1
SO-1,6
BL-2
	c.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		`class Student {`
		`  private int sid;`
		`  private String name;`
		`  private Student() {`
		`  sid = 101;`
		`  name = "Test";`
		`  }`
		`  void printInfo() {`
		`  System.out.println("S-ID: " + sid);`
		`  System.out.println("Name: " + name);`
		`  }`
		`}`
			[02]

		public class Main {
		public static void main(String[] args) {
		Student S = new Student();
		S.printInfo();
		}
		}	

CO1 SO-1,6 BL-2	d.	Explain JDK in detail.	[02]
		Answer any three questions from Q2 to Q5 (Each question is for four marks)	
Q2 CO1 SO-1,6 BL-2		Explain features of Java (any 8)	[04]
Q3 CO1 SO-1,6 BL-2		WAP to accept a two digit number from the user using BufferedReader class and display the two digits separately.	[04]
Q4 CO1 SO-1,6 BL-4		WAP to display the following pattern * * * * * * * * * * . . n lines	[04]
Q5 CO1 SO-1,6 BL-4		WAP to count the number of objects made of a particular class using static variable and static method to display the same.	[04]

All The Best!!!

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SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA

Subject: Java Programming

Date: 23 / 08 / 2023

Marks: 20

Year: I Semester: I

Time: 45 mins

No. of Pages: 02

Midterm Test – I (Set A)

Instructions:
1) Q1 is compulsory. Attempt any three questions from the remaining.
2) Output is mandatory for all programming questions.

Q1		Answer briefly:	[08]
CO1 SO-1,6 BL-2	a.	Explain Classes and Objects.	[02]

CO1
SO-1,6
BL-3
	b.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		`public class Main {`
		`  public static void main(String[] args) {`
		`  int a = 10;`
		`  int b = 20;`
		`  int c = 0;`
		`  c = ++a + ++b * 2 + a++;`
		`  System.out.println(a + ", + b + "," + c);`
		`  }`
		`}`
			[02]

CO1
SO-1,6
BL-2
	c.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		`class Student {`
		`  private int sid;`
		`  private String name;`
		`  Student(int id, String na) {`
		`  sid = id;`
		`  name = na;`
		`  }`
		`  void printInfo() {`
		`  System.out.println("S-ID: " + sid);`
		`  System.out.println("Name: " + name);`
		`  }`
		`}`
			[02]

		public class Main {
		public static void main(String[] args) {
		Student S1 = new Student();
		Student S2 = new Student(101, "Rohit");
		
		S1.printInfo();
		S2.printInfo();
		}
		}	

CO1 SO-1,6 BL-2	d.	Explain JRE in detail.	[02]
		Answer any three questions from Q2 to Q5 (Each question is for four marks)	
Q2 CO1 SO-1,6 BL-2		Explain Constructor, its types and rules for creating constructors.	[04]
Q3 CO1 SO-1,6 BL-2		WAP to find largest of three numbers accepted from command line.	[04]
Q4 CO1 SO-1,6 BL-4		WAP to display the following pattern A B C D E F G H I J . . n lines	[04]
Q5 CO1 SO-1,6 BL-4		WAP to demonstrate constructor overloading for finding area of rectangles and displaying it.	[04]

All The Best!!!

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA | Year: I | Semester: I

Subject: Java Programming | Time: 45 mins

Date: 04 / 10 / 2023 | No. of Pages: 02

Marks: 20

Midterm Test – II (Set A)

1) Q1 is compulsory. Attempt any three questions from the remaining.
2) Answer to each new question to be started on a fresh page.
3) Figures in brackets on the right hand side indicate full marks.
4) Assume Suitable data if necessary.
5) Output is mandatory for all programming questions.

Q1		Answer briefly:	[08]
CO1 SO-1,6 BL-2	a.	Explain Hierarchical Inheritance with an example.	[02]

CO1
SO-1,6
BL-3
	b.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		`interface A`
		`{`
		`  int i = 111;`
		`}`
		`class B implements A`
		`{`
		`  void methodB()`
		`  {`
		`  i = 222;`
		`  System.out.println(i);`
		`  }`
		`}`
		`class Main`
		`{`
		`  public static void main(String args[])`
		`  {`
		`  B s=new B();`
		`  s.methodB();`
		`  }`
		`}`
			[02]

CO1
SO-1,6
BL-2
	c.	How many string objects are created in the following code? Justify your answer.
		`String s1 = "Helloworld";`
		`String s2 = "Helloworld";`
		`String s3 = "Greeting";`
		`String s4 = new String("Greeting");`
		`String s5 = "Helloworld";`
			[02]

CO1 SO-1,6 BL-2	d.	Explain compile-time Exceptions in detail.	[02]
		Answer any three questions from Q2 to Q5 (Each question is for four marks)	
Q2 CO1 SO-1,6 BL-2		Differentiate between Abstract class and Interface.	[04]
Q3 CO1 SO-1,6 BL-2		Write a program to calculate the total number of characters in the String excluding blank spaces? Eg: input string: Hello World No. of characters : 10	[04]
Q4 CO1 SO-1,6 BL-4		WAP to handle ArrayIndexOutOfBoundsException using command line arguments input and try catch finally block.	[04]
Q5 CO1 SO-1,6 BL-4		Explain the steps to create and run a package in java. Support your answer with a code.	[04]

All The Best!!!

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA | Year: I | Semester: I

Subject: Java Programming | Time: 45 mins

Date: 04 / 10 / 2023 | No. of Pages: 02

Marks: 20

Midterm Test – II (Set C)

1) Q1 is compulsory. Attempt any three questions from the remaining.
2) Answer to each new question to be started on a fresh page.
3) Figures in brackets on the right hand side indicate full marks.
4) Assume Suitable data if necessary.
5) *Output is mandatory for all programming questions.*

Q1		Answer briefly:	[08]
CO1 SO-1,6 BL-2	a.	Explain super keyword in detail.	[02]

CO1
SO-1,6
BL-3
	b.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		
		`class A`
		`{`
		`  int i = 111;`
		`}`
		
		`interface B extends A`
		`{`
		`  void methodB()`
		`  {`
		`  System.out.println(i);`
		`  }`
		`}`
		`class Main`
		`{`
		`  public static void main(String args[])`
		`  {`
		`  B s=new B();`
		`  s.methodB();`
		`  }`
		`}`
			[02]

CO1
SO-1,6
BL-2
	c.	Determine if the following code will compile? If not, justify why? If yes, write the output.
		`public class Main {`
		`  public static void main(String[] args)`
		`  {`
		`  String s1 = new String("HELLO");`
		`  String s2 = new String("HELLO");`
		`  System.out.println(s1 == s2);`
		`  System.out.println(s1.equals(s2));`
		`  }`
		`}`
			[02]

CO1 SO-1,6 BL-2	d.	Explain StringBuffer class in detail.	[02]
		Answer any three questions from Q2 to Q5 (Each question is for four marks)	
Q2 CO1 SO-1,6 BL-2		Explain Abstract Class and Abstract Method with an example code.	[04]
Q3 CO1 SO-1,6 BL-2		WAP to perform following String operations **using String methods**. Take two string inputs from the user. Eg: string 1 : Hello World !!! and String 2 : Hello. Perform the following string operations on it. 1. Concatenation of string 1 and string 2. 2. Determine if string 2 occurs in string 1. If string 2 occurs in string 1, then extract string 2 from string 1. 3. replace string 2 by "Java" in string 1. 4. compare string 1 and string 2 using the method that returns integer value after comparing.	[04]
Q4 CO1 SO-1,6 BL-4		WAP to handle Arithmetic Exception and Number Format exception using try catch finally block.	[04]
Q5 CO1 SO-1,6 BL-4		WAP to create a sub package inside a package and also show the execution of the same.	[04]

All The Best!!!

MPSTME, NMIMS University
MCA, Semester -I, AY 2023-24

Test – 2 (Set – 1)
Subject: Data Structures and Algorithms

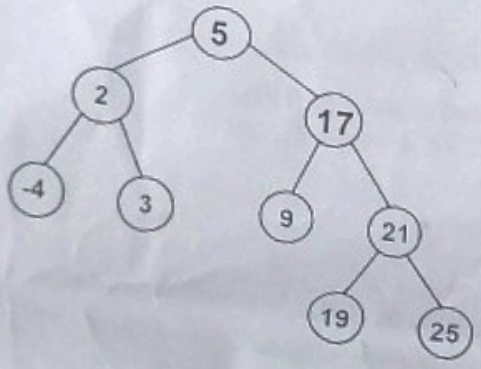
Time: 50 min.
Marks: 20

Q1. Solve the following questions

a. Explain the concept of doubly liked list with application and advantages. [03]
b. Given a sequence of values 20, 14, 30, 17, 10, 16, 22, 40, 5. Generate Binary Search Tree for the given sequence and display "inorder" traversal sequence [03]
c. State the significance of dynamic memory allocation with suitable example [02]

Q2. Solve any two from the following (all questions carries equal marks) [12]

a. Discuss the suitable application of stack where linked list implementation is necessary. Perform push and pop operations with suitable algorithm on stack using linked list.
b. Perform insert operation and generate complete binary tree for the data sequence: 12, 34, 56, 7, 15, 66, 78, 34, 45, 74, 3, 19, 98. Produce "postorder" sequence with the help of algorithm from the generated tree.
c. Perform delete operation on following BST. Identify the cases for deletion and discuss with suitable algorithms.



MPSTME, NMIMS University
MCA, Semester -I, AY 2023-24

Test – 1 (Set – 2) Time: 50 min.
Subject: Data Structures and Algorithms Marks: 20

Q1. Solve the following questions

a. Given an array as 12,14,16,17,18,19. Write algorithm / program to insert 20 at index 3, index starts at 0, also display all the elements in sequence after insertion. [03]
b. What is the significance of asymptotic notations? Explain with suitable example [03]
c. Explain abstract data types with suitable example [02]

Q2. Solve any two from the following (all questions carries equal marks) [12]

a. In a waiting line people enter sequentially at random time and leave at random time to get certain service. Only 6 chairs are available in the line to occupy at any point of time. At least 15 people will get the service done, only once in sequence.
Identify suitable data structure to implement above-mentioned scenario and explain the operations required with algorithm / program to provide service to every individual.
b. Compiler read the mathematical expression in infix notation and evaluate by converting it into either prefix or postfixed expression. Example of the expression is given below.
Example: x = (a + (b – c) * d) + (e * 2)
Using suitable data structure convert and evaluate above expression, also indicate every step required.
c. Discuss the factors considered in calculating space and time complexity. State the significance of looping in considering time complexity with appropriate example.

MPSTME, NMIMS University
MCA, Semester -I, AY 2023-24

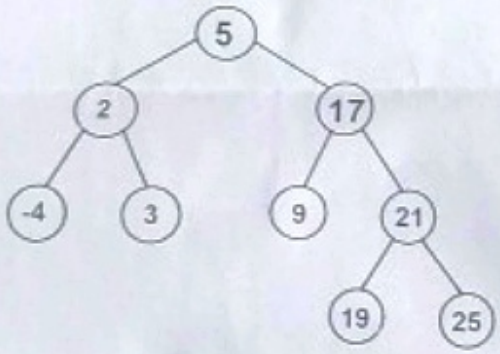
Test – 2 (Set – 1) Time: 50 min.
Subject: Data Structures and Algorithms Marks: 20

Q1. Solve the following questions

a. Explain the concept of doubly liked list with application and advantages. [03]
b. Given a sequence of values 20, 14, 30, 17, 10, 16, 22, 40, 5. Generate Binary Search Tree for the given sequence and display "inorder" traversal sequence [03]
c. State the significance of dynamic memory allocation with suitable example [02]

Q2. Solve any two from the following (all questions carries equal marks) [12]

a. Discuss the suitable application of stack where linked list implementation is necessary. Perform push and pop operations with suitable algorithm on stack using linked list.
b. Perform insert operation and generate complete binary tree for the data sequence: 12, 34, 56, 7, 15, 66, 78, 34, 45, 74, 3, 19, 98. Produce "postorder" sequence with the help of algorithm from the generated tree.
c. Perform delete operation on following BST. Identify the cases for deletion and discuss with suitable algorithms.



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream : Computer Year: I Semester: II

Subject: Computer Networks Time: 45 Minutes(5:00 to 5:45)

Date: 22.08.2023 No. of Pages: 2

Marks: 20

Test-I

SET B

Instructions: Candidates should read carefully the instructions.
1) **Answer to each new question to be started on a fresh page.**
2) **Figures in brackets on the right hand side indicate full marks.**
3) **Assume Suitable data if necessary.**
4) **Question no.1 is compulsory.**
5) **Solve any three questions from the remaining questions.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly (Each question is for two marks):</td><td>[08]</td></tr>
<tr><td>CO-1 ;
SO- 1;
BL-2</td><td>a.</td><td>Match the following based on the network Topologies.
<table>
<tr><td>1. Topology has a central hub that connects all devices in the network</td><td>A. Bus Topology</td></tr>
<tr><td>2. Each device is connected to exactly two other devices, creating a circular arrangement.</td><td>B. Star Topology</td></tr>
<tr><td>3. Devices are connected linearly in this topology, with each device having a dedicated connection to the next device.</td><td>C. Ring Topology</td></tr>
<tr><td>4. Topology provides a balance between the reliability of a ring topology and the scalability of a star topology</td><td>D. Hybrid Topology</td></tr>
</table></td><td>[02]</td></tr>
<tr><td>CO-1 ;
SO- 1;
BL-1</td><td>b.</td><td>Define WAN.</td><td>[02]</td></tr>
<tr><td>CO-1 ;
SO-1 ;
BL-1</td><td>c.</td><td>State the difference between Half Duplex and Full Duplex transmission modes.</td><td>[02]</td></tr>
</table>

CO-2 ; SO-2 ; BL-2	d.	What is the sender and receiver window size in Stop and Wait ARQ. Explain your answer.	[02]
Answer any three questions from Q2 to Q5 (Each question is for four marks)			
Q2 CO-1 ; SO-1; BL-4		Explain Star Topology. If you have a network with 8 devices in a star topology, calculate the total number of ports required for the central hub and each individual device. How many total cables would be needed to connect all devices to a central hub?	[04]
Q3 CO-2; SO-2; BL-4		Draw the flow diagram for Go Back N ARQ. Explain the following cases in brief: 1. Ideal case (No frame or acknowledgment lost) 2. Frame 1 is lost.	[04]
Q4 CO-1;SO-1; BL-4		Explain the responsibilities of Network Layer & Transport layer in OSI model.	[04]
Q5 CO-1;SO-1; BL-4		Why is Optical Fiber cable superior to Twisted pair cable? Explain Optical Fiber cable is detail.	[04]

All The Best!!!

Physic

Session
Transport
Network
Data link
physical

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream : Computer | Year: I Semester: I

Subject: Computer Networks | Time: 45 Minutes(3:00 to 3:45)

Date: 05.09.2023 | No. of Pages: 2

Marks: 20

Test-II

SET B

Instructions: Candidates should read carefully the instructions.
1) **Answer to each new question to be started on a fresh page.**
2) **Figures in brackets on the right hand side indicate full marks.**
3) **Assume Suitable data if necessary.**
4) **Question no.1 is compulsory.**
5) ***Solve any three questions from the remaining questions.***

Q1		Answer briefly (Each question is for two marks):	[08]
CO-2 ; SO-6 ; BL-3	a.	Compared to other flow control algorithm 'selective repeat ARQ' has a window size more the 1. How it improves the flow control mechanism. Explain.	[02]
CO-3 ; SO-1 ; BL-3	b.	Given the IP address 11111001 10011011 11111011 00001111, identify the class and find the network address (in decimal format).	[02]
CO-2 ; SO-1,6 ; BL-3	c.	In a even parity hamming code. Dataword is given as 1010101011101. How many parity bits are required to form the hamming code. (Also mention the formula)	[02]
CO-4 ; SO-6; BL-2	d.	Explain the following from IPv4 packet format: 1. Protocol 2. Type of service Mention the bits allotted.	[02]
Answer any three questions from Q2 to Q5 (Each question is for four marks)			
Q2 CO-2 ; SO-1,6; BL-2		Explain Aloha with the help of suitable diagram.	[04]

Q3 CO-2; SO-1,6; BL-6		A sender wants to transmit a message 1101011010 to a receiver using CRC with a generator polynomial of x^3+x+1. Perform CRC encoding and provide the transmitted message. Show all intermediate steps in your calculation.	[04]
Q4 CO3; SO-1; BL-2		Explain link state routing algorithm with suitable diagrams.	[04]
Q5 CO-3;SO-1; BL-2		Explain datagram switching with suitable diagram	[04]

All The Best!!!

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream : Computer Year: I Semester: I

Subject: Computer Networks Time: 45 Minutes(3:00 to 3:45)

Date: 05.09.2023 No. of Pages: 2

Marks: 20

Test-II

SET C

Instructions: Candidates should read carefully the instructions.
1) **Answer to each new question to be started on a fresh page.**
2) **Figures in brackets on the right hand side indicate full marks.**
3) **Assume Suitable data if necessary.**
4) **Question no.1 is compulsory.**
5) **Solve any three questions from the remaining questions.**

Q1	Answer briefly (Each question is for two marks):		[08]
CO-2 ; SO-6 ; BL-3	a.	Write the sender side window size for Selective Repeat ARQ where m=9.	[02]
CO-3 ; SO-1 ; BL-3	b.	Given the IP address 01001011 00101101 00100010 01001110, identify the class and find the network address (in decimal format).	[02]
CO-2 ; SO-1,6 ; BL-3	c.	Let us assume that dataword is of 2 bits , and codeword for 3 bits. List all the possible datawords and codewords possible for this combination. State Valid and Invalid.	[02]
CO-4 ; SO-6; BL-2	d.	Explain the following from IPv4 packet format: 1. Flag 2. Identification Mention the bits allotted.	[02]
Answer any three questions from Q2 to Q5 (Each question is for four marks)			
Q2 CO-2 ; SO-1,6; BL-2		Explain CSMA/CA with the help of suitable diagram.	[04]

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Q3 CO-2; SO-1,6; BL-6		Encode a binary word 11001 into the even parity hamming code.	[04]
Q4 CO3; SO-1; BL-2		Explain count to infinity problem in detail with suitable examples.	[04]
Q5 CO-3;SO-1; BL-2		Explain virtual circuit switching with suitable diagram	[04]

All The Best!!!

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream : Computer

Year: I Semester: II

Subject: Computer Networks

Time: 45 Minutes(5:00 to 5:45)

Date: 22.08.2023

No. of Pages: 2

Marks: 20

Test-I

SET C

Instructions: Candidates should read carefully the instructions.
1) **Answer to each new question to be started on a fresh page.**
2) **Figures in brackets on the right hand side indicate full marks.**
3) **Assume Suitable data if necessary.**
4) **Question no.1 is compulsory.**
5) **Solve any three questions from the remaining questions.**

<table>
<tr><td>Q1</td><td colspan="2">Answer briefly (Each question is for two marks):</td><td>[08]</td></tr>
<tr><td>CO-1 ; SO- 1; BL-2</td><td>a.</td><td>Match the following based on TCP/IP Model.
<table>
<tr><td>1. Provides access to the network for the end user</td><td>A. Network Layer</td></tr>
<tr><td>2. End to end Flow Control</td><td>B. Application Layer</td></tr>
<tr><td>3. Route Determination</td><td>C. Transport Layer</td></tr>
<tr><td>4. Interface to physical world</td><td>D. Physical Layer</td></tr>
</table></td><td>[02]</td></tr>
<tr><td>CO-1 ; SO-1 ; BL-1</td><td>b.</td><td>Compare LAN & WAN.</td><td>[02]</td></tr>
<tr><td>CO1- ; SO-1 ; BL-1</td><td>c.</td><td>Explain the different types of connection: Point to Point and Multipoint.</td><td>[02]</td></tr>
<tr><td>CO- 2; SO-2 ; BL-2</td><td>d.</td><td>What is the sender window size in Go Back N ARQ, if m = 4 bits? Explain your answer.</td><td>[02]</td></tr>
</table>

Answer any three questions from Q2 to Q5 (Each question is for four marks)			
Q2 CO-1 ; SO-1; BL-4		Explain Mesh Topology. Assume six devices are arranged in a mesh topology. how many cables are needed? How many ports are required for each device?	[04]
Q3 CO-2; SO-2; BL-4		Explain what will happen if window size of sender side in Go Back N ARQ is 2^m instead of 2^m-1. Draw suitable diagram.	[04]
Q4 CO-1;SO-1; BL-4		Explain the responsibilities of Data Link Layer & Presentation Layer in OSI model.	[04]
Q5 CO-1;SO-1; BL-4		Explain the three modes of propagation in wireless media. Which wave is omnidirectional in nature, explain.	[04]

All The Best!!!

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream :Computer Year: I Semester: I

Subject: DBMS Time: 45 Minutes. (3 to 3:45)

Date: 28/08/2023

Marks: 20

Test-I (Set A)

Q1	Answer briefly (Each question is for two marks):		[08]
	a.	Explain Functions of DBMS	2M
	b.	Explain the concept of referential integrity with the help of an example	2M
	c.	Give the advantage of using DBMS over file System	2M
	d.	Explain Transaction Management	2M
Q2		Consider a MAIL_ORDER database in which employees take orders for parts from customers. The data requirements are summarized as follows: • The mail order company has employees, each identified by a unique employee number, first and last name, and Zip Code. • A unique customer number, first and last name, and Zip Code identify each customer of the company. • A unique part number, a part name, price, and quantity in stock identify each part sold by the company. • Each order placed by a customer is taken by an employee and is given a unique order number. • Each order contains specified quantities of one or more parts. • Each order has a date of receipt as well as an expected ship date. The actual ship date is also recorded. **a) Draw ER diagram according to the above requirements.** **b) Convert the ER diagram into equivalent schema**	[08]
Q3		Discuss the rules for converting ER Model to Relation Model for the following with the help of an example 1. Composite Attribute 2. One to Many relationship 3. Many to Many relationship 4. Unary relationship	[04]

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream :Computer Year: I Semester: I

Subject: DBMS Time: 45 Minutes. (3 to 3:45)

Date: 28/08/2023

Marks: 20

Test-I (Set A)

Q1	Answer briefly (Each question is for two marks):		[08]
	a.	Explain Functions of DBMS	2M
	b.	Explain the concept of referential integrity with the help of an example	2M
	c.	Give the advantage of using DBMS over file System	2M
	d.	Explain Transaction Management	2M
Q2		Consider a MAIL_ORDER database in which employees take orders for parts from customers. The data requirements are summarized as follows: • The mail order company has employees, each identified by a unique employee number, first and last name, and Zip Code. • A unique customer number, first and last name, and Zip Code identify each customer of the company. • A unique part number, a part name, price, and quantity in stock identify each part sold by the company. • Each order placed by a customer is taken by an employee and is given a unique order number. • Each order contains specified quantities of one or more parts. • Each order has a date of receipt as well as an expected ship date. The actual ship date is also recorded. **a) Draw ER diagram according to the above requirements.** **b) Convert the ER diagram into equivalent schema**	[08]
Q3		Discuss the rules for converting ER Model to Relation Model for the following with the help of an example 1. Composite Attribute 2. One to Many relationship 3. Many to Many relationship 4. Unary relationship	[04]

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MCA Stream :Computer Year: I Semester: I

Subject: DBMS Time: 45 Minutes. (3 to 3:45)

Date: 9/10/23

Marks: 20

Test-I (Set A)

Q1	Answer briefly (Each question is for two marks):		[08]
	a.	Explain 2 DDL Commands with the help of an example	2M
	b.	Write syntax for adding check and unique constraint in table using alter command	2M
	c.	Why do we need normalization?	2M
	d.	What is the Condition to check 2nd Normal Form explain with example	2M
Q2		**Write SQL Queries for given questions based on below tables** **a. Find the total number of courses offered.** **b. Find the courses taught by "Professor Smith."** **c. Retrieve a list of all students and their respective course enrollments (if any).** **d. Calculate the total number of students enrolled in each course, and list only courses with more than three students.** **e. Retrieve the student name who is enrolled in the "Computer Science" course.**	[08]
Q3		Find all the keys for given relation and give prime and non-prime attribute. Explain any one rule of Functional Dependency **R (ABCDE)** FD= {A->B, BC->E and ED->A } **R (ABCDEF)** FD = {CDE->B, ACD->F, BEF-> C and B-D }	[04]

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

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Subject: DBMS Time: 45 Minutes. (3 to 3:45)

Date: 9/10/23

Marks: 20

Test-I (Set B)

Q1	Answer briefly (Each question is for two marks):		[08]
	a.	Explain 2 DML Commands with the help of an example	2M
	b.	Write syntax for removing foreign key constraint from table using alter command	2M
	c.	What is Insert and Update Anomaly? Explain with example	2M
	d.	What is the condition to check 3rd and BCNF explain with example	2M
Q2		**Write SQL Queries for given questions based on below tables** **a. Find the name of student whose first name starts with 'J'** **b. Give the details of the Faculty who teaches 'Computer Science'** **c. Retrieve the first and last names of students who are not enrolled in any course.** **d. List the courses along with the count of students enrolled in each course. Include courses with zero students.** **e. Find the names of students who are enrolled in the "Mathematics" course**	[08]
Q3		Find all the keys for given relation and give prime and non-prime attribute. Explain any one rule of Functional Dependency **R (ABCDE)** FD= {B->C, B->E, CD->B and DE -> C} **R (ABCDEFG)** FD = {AB->C, CD->E, EF->G, FG->E, DE->C and BC->A}	[04]

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-24

Program: MCA Stream: Computer Applications

Year:1st Semester: I

Subject: OS

Time: 45 Minutes. (10.00 to 10.45)

Date: 29-08-23

No. of Pages: 1

Marks: 20

Test-I [SET-B]

Instructions: Candidates should read the instructions carefully.
1) Answer to each new question to be started on a fresh page.
2) Figures in brackets on the right-hand side indicate full marks.
3) Assume Suitable data if necessary and draw diagrams wherever necessary.
4) Q1 is compulsory. Attempt anyone from Q2 to Q5.

Q1		Answer briefly:	
CO- 1; SO-4; BL-1	a.	Describe Multiprocessor systems and categorize their types and advantages	02
CO-1; SO-4; BL-2	b.	Differentiate User mode vs. Kernel mode.	02
CO- 2; SO-4; BL-3	c.	Discuss the structure of Thread and list its types	02
CO- 1; SO-4; BL-2	*d.*	*List the various services of OS. (User and System perspective).*	02
Q2 CO-1; SO-5; BL-4		Explain Microkernel architecture and Identify 2 Advantages and 2 Disadvantages.	04
Q3 CO-2 ; SO-5; BL-3		For the given table, which of the following scheduling algorithms (FCFS, Preemptive Priority) will work best in the case of average waiting time? Require Gantt chart and full steps.	04
Q4 CO-1;SO-4; BL-4;		Explain the different Computing Environments.	04
Q5 CO-1;SO-5; BL-5		Processes are heavyweight, and Threads are lightweight. Justify	04

Process Name	Arrival Time	Burst Time	Priority
1	0	5	3
2	1	7	1
3	3	4	2

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-24

Program: MCA Stream: Computer Applications Year:1st Semester: 1

Subject: OS Time: 45 Minutes. 10.15 to 11

Date: 03-10-23 No. of Pages: 1

Marks: 20

Test-I [SET-A]

Instructions: Candidates should read the instructions carefully.
1) Answer each new question to be started on a fresh page.
2) Figures in brackets on the right-hand side indicate full marks.
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4) Q1 is compulsory. Any three from Q2 to Q5

Q1 CO-1, SO-3,BL-4	a.	Differentiate Co-operative vs. Concurrent processes.	2
CO-1,SO-3,BL-2	b.	Discuss the Deadlock prevention with respect to "Hold and Wait" and state its 3 disadvantages.	2
CO-1,SO-4,BL-1	c.	List out the levels of process queues in Multilevel queue scheduling	2
CO-1,SO-2,BL-2	d.	Discuss the role of "Disabling interrupts" in solving the Critical section.	2
Q2 CO-1,SO-2,BL-2		Explain the Midterm scheduling with the concept of context-switching	4
Q3 CO-2,SO-4,BL-4		Consider the following set of processes with the length of the CPU burst time given in milliseconds PROCESS / AT / BT P1 / 0 / 7 P2 / 2 / 4 P3 / 3 / 2 P4 / 9 / 1 Draw a Gantt chart for RR-2 and calculate the average waiting time.	4
Q4 CO-2,SO-4,BL-1		Write "Safety Algorithm" for the deadlock avoidance	4
Q5 CO-1,SO-3,BL-1		Define the terms 1) Race condition 2) RAG graph	4

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-24

Program: MCA Stream: Computer Applications Year:1st Semester: 1

Subject: OS Time: 45 Minutes. 10.15 to 11

Date: 03-10-23 No. of Pages: 1

Marks: 20

Test-I [SET-A]

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SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-24

Program: MCA Stream: Computer Applications Year:1st Semester: I

Subject: OS Time: 45 Minutes. (10.00 to 10.45)

Date: 29-08-23 No. of Pages: 1

Marks: 20

Test-I [SET-B]

Instructions: Candidates should read the instructions carefully.
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CO- 2; SO-4; BL-3	c.	Discuss the structure of Thread and list its types	02
CO- 1; SO-4; BL-2	d.	List the various services of OS. (User and System perspective).	02
Q2 CO-1; SO-5; BL-4		Explain Microkernel architecture and Identify 2 Advantages and 2 Disadvantages.	04
Q3 CO-2 ; SO-5; BL-3		For the given table, which of the following scheduling algorithms (FCFS, Preemptive Priority) will work best in the case of average waiting time? Require Gantt chart and full steps.	04

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